

DSML 2026: Dependable and Secure Machine Learning

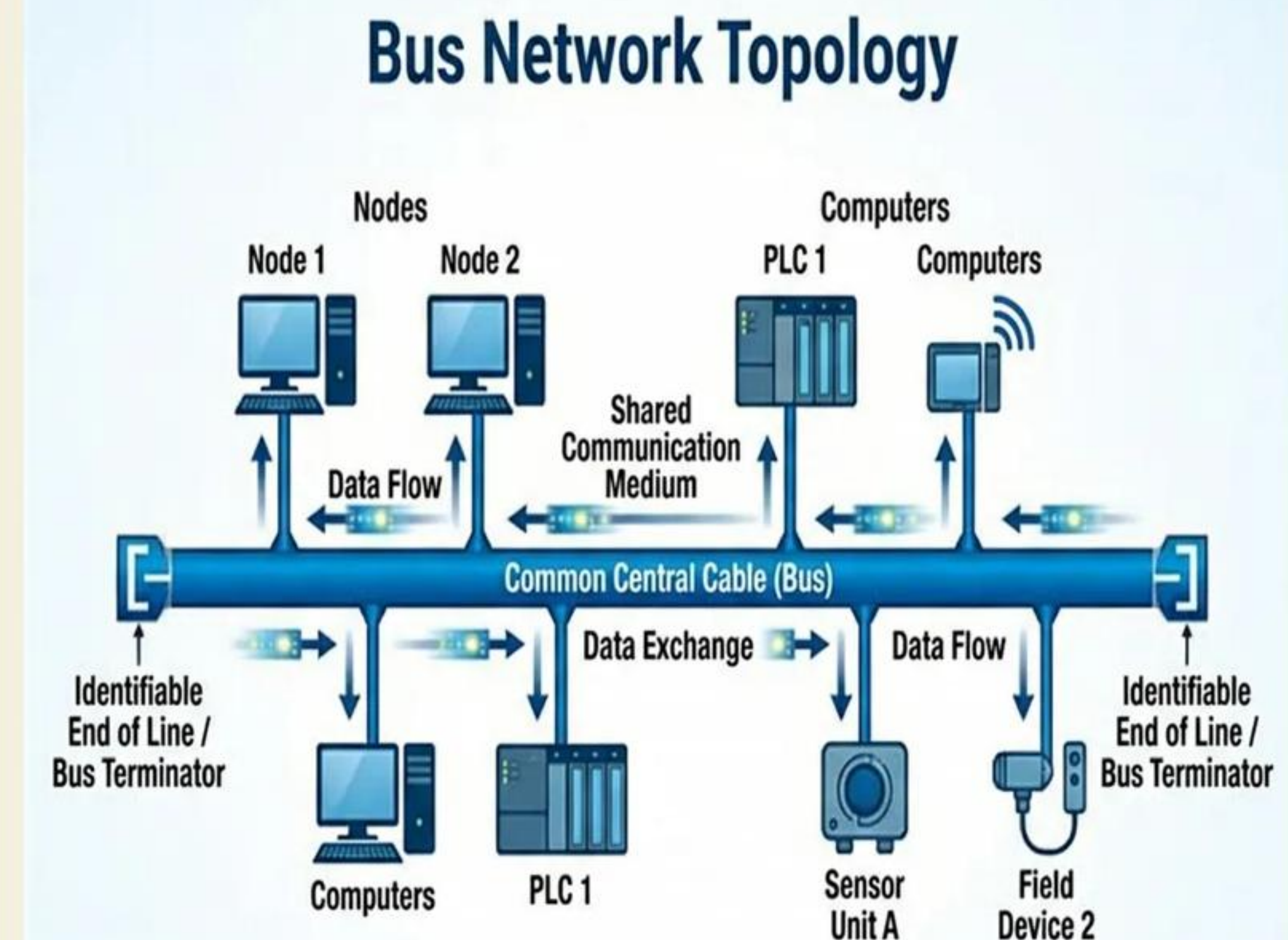
Establishing Zero-Shot LLM Performance at Network Tomography

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Motivation: The Anonymisation Gap

- > **Network Tomography:** Reconstructing internal structure from external observations (NetFlow).
- > **The Paradox:** Topology awareness aids management and synthetic data but introduces massive privacy risks.
- > **Anonymisation Failure:** Conventional metadata removal ignores high-dimensional statistical patterns.
- > **The LLM Threat:** Zero-shot pattern recognition allows "Invisible Reconnaissance" without active probing.



Primary Research Inquiries



RQ1: Baseline

What is the baseline capability of LLMs for generating syntactically valid NetFlow data?



RQ2: Inference

Are LLMs able to infer structural and topological features from limited NetFlow samples?



RQ3: Training Data

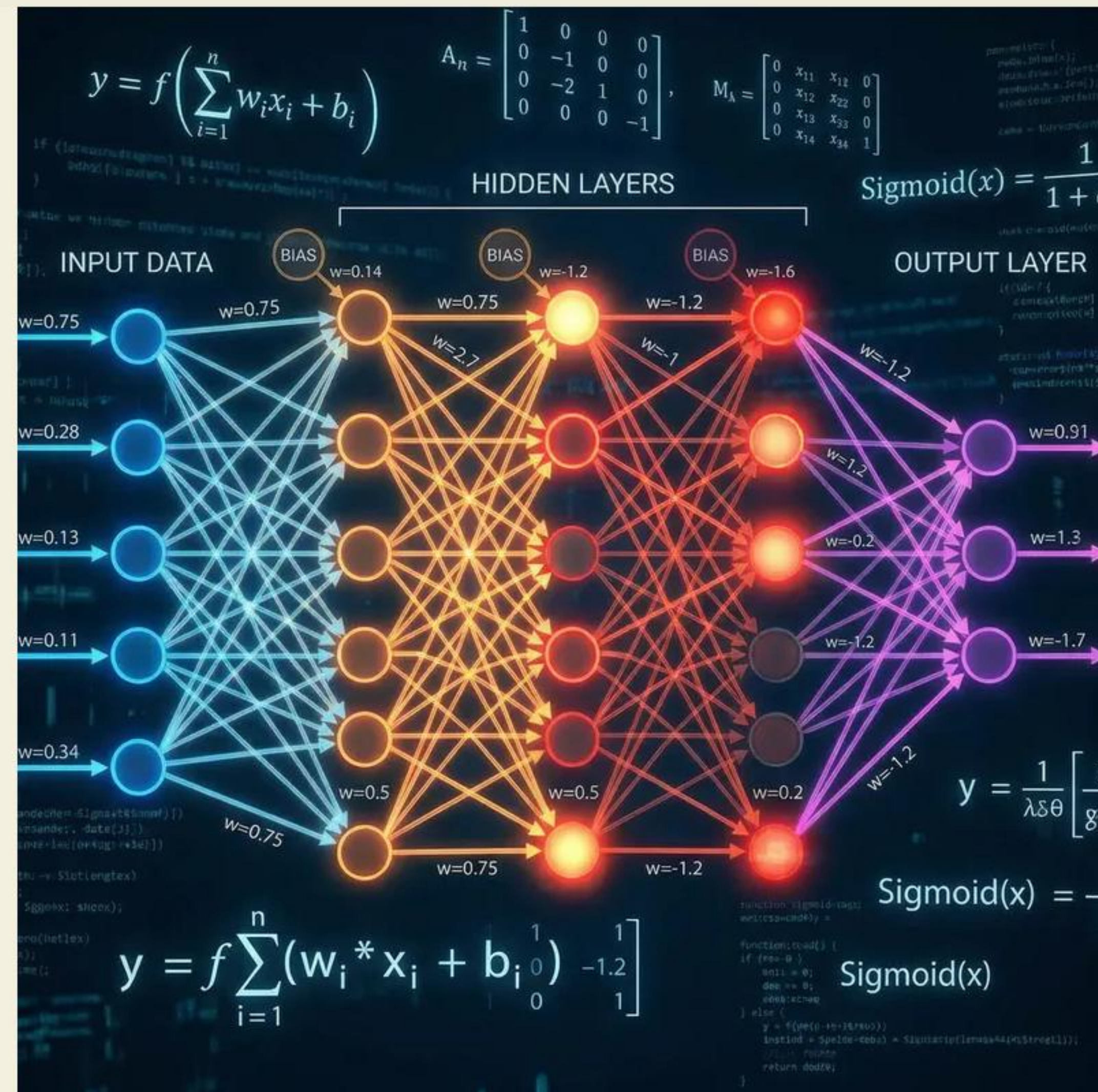
How does the inclusion of structured data (code/math) impact topological reasoning?



RQ4: Data Types

Do LLMs exhibit distinct ability differences between static and dynamic data types?

Experimental Setup: Models Evaluated



14 Diverse LLM Architectures

- > **Open-Weight (6–9B):** LLaMa2, CodeLLaMa, LLaMa3.1, Tulu3, Qwen2.5, Qwen2.5-Coder, Gemma, CodeGemma, DeepSeek Family.
- > **Frontier Benchmark:** Gemini-2.0-Flash.
- > **Categories:** Foundation Models vs. Code-Tuned variants vs. 'From-Scratch' specialized models.
- > **Framework:** Ollama for generation; identical prompts and JSON constraints.

NetFlow Infilling Task

The Dataset

Realistic simulated NetFlow provided by BT, generated via **Keysight Ixia's BreakingPoint**.

Simulates video streaming, corporate traffic, and standard routing across 48 distinct fields.

Sequence Completion

- > Input: 5, 10, 20, or 50 complete NetFlow records.
- > Task: Infill a "damaged" record with masked fields (Start Removed, End Removed, or Entire Record).
- > Evaluation: 100 distinct inputs per scenario.

Measuring Topological Understanding

- **Possible Score:** Percentage of syntactically valid and in-range values.
- **Mean Partial Match (MPM):** Ratio of correct octets in generated IP addresses.
- **Total Variation Complement (TVC):** Distributional similarity for categorical fields.
- **Correct Type (CT):** Mapping values to semantic groups (e.g., IANA port ranges).

 Evaluation metrics and charts

Categorical Inference Success

Protocol & Flags

Models learned the rigid rule: **TCP flags only apply to TCP protocols.**

High accuracy for *Protocol* (PR) and *Type of Service* (TOS) across most foundation and code-tuned models.

Port Category Mapping

Destination ports were significantly more predictable than source ports.

Models grasped semantic categories (System vs Register ports) even when missing exact values.

Finding: LLMs infer underlying network rules, not just random tokens.

Subnet and IP Reconstruction

Model	SA Octet 01	SA Octet 02	SA Octet 03	Entire IP (All)
CodeLLaMa	86	86	49	6
LLaMa3.1	67	67	36	8
Tulu3	97	97	48	10
Gemma	19	19	6	0
Gemini-2.0-Flash	71	71	41	2

Insight: High MPM scores for the first three octets suggest models effectively place hosts in the correct subnets, demonstrating logical network awareness.

The Capability Gap: Routing Logic

Frontier Model Performance

- Gemini-2.0-Flash demonstrated sophisticated routing inference.
- **89% Accuracy** for Output Interface when IPs were provided.
- Open-source models failed to capitalize on partial context for routing.

Path Construction (Correct Values)



The Boundary: Numerical Collapse

Performance Collapse in Dynamic Fields

While structural inference is strong, models struggle with dynamic, unbounded numerical features:

- > **Flow Size (ibyt, obyt):** Median percentage difference frequently **-200%**.
- > **Flow Duration:** Failure to model logical dependencies with timestamps.
- > **Synthetic Fidelity:** Current zero-shot models cannot generate realistic traffic volumes.

Median \% Difference $\approx -200\%$ \text{ for Flow Volumes}

Context Window Dynamics

Optimal Threshold

Performance typically peaks at **10 to 20 rows** of context. Increasing context beyond this often leads to performance decay due to noise.

Frontier Resilience

Gemini-2.0-Flash maintains high performance at 50 rows, unlike smaller models. Smaller models exhibit a collapse at 50 rows, likely due to training sequence length limits.

Conclusion & Security Risks



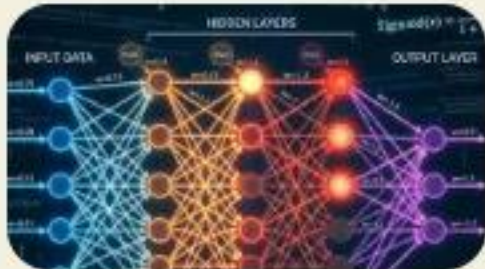
- > **Information Leakage:** NetFlow data, though anonymized, allows detailed reconnaissance via LLMs.
- > **CWE-200:** Classified as "Exposure of Sensitive Information to Unauthorized Actor".
- > **DREAD Risk:** Overall score of **8.8 / 10**.
- > **Future Work:** Domain-specific fine-tuning and LLM-resistant obfuscation techniques.

Image Sources



https://media.easy-peasy.ai/b93d84f1-6de6-491b-aa51-4daa8a3a85f2/ff62788b-ccd9-4ccd-8b87-ceed16b881d8_medium.webp

Source: easy-peasy.ai



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Source: easy-peasy.ai



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